

# Enterprise Network Infrastructure & Active Directory Deployment

Multi-Site Network Design, Windows Server Administration & Cloud Migration  
Strategy for a Two-Office Enterprise (Johannesburg & Pretoria)

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COMPETENCY LEVEL

Associate — Network & Systems Administration

IP Addressing & VLSM

Hierarchical Switching

Spanning Tree Protocol

Windows Server 2016

Active Directory

Group Policy

File & Print Services

Microsoft Azure

01 PROJECT OVERVIEW

Objective

This portfolio documents the end-to-end design and implementation of a converged network and systems infrastructure for VTG (Virtual Technology Group), a two-site enterprise operating branch offices in Johannesburg and Pretoria. The brief required a single, coherent infrastructure solution spanning four disciplines: structured IP address planning and subnetting, hierarchical switch topology design with loop-prevention, Windows Server-based Active Directory administration (organisational units, users, groups, file services and security policy), and a forward-looking cloud migration strategy built on Microsoft Azure.

The objective throughout was to treat the two offices as a single logical enterprise rather than two isolated sites — a consistent addressing scheme, a shared domain namespace, and a unified security baseline were applied across both locations so that the resulting design could scale, be centrally managed, and support remote and hybrid working patterns.

Scope & Context

The infrastructure was built and validated inside a virtualised Windows Server 2016 lab environment (domain **TiisetsoHVTG.com**), representing the production environment VTG would deploy. Two domain controllers were provisioned — one representing the Johannesburg site and one representing Pretoria — with department-level structure, addressing, and policy mirrored across both in line with the network design produced in the planning phase.

Deliverables Covered in this Portfolio

|                        |                                                                                                                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Design         | Subnetted IP address plan (VLSM) for five departments per site across LAN and wireless segments; three-tier hierarchical switch topology (access · distribution · core); Spanning Tree Protocol strategy for loop prevention. |
| Directory Services     | Organisational Unit (OU) structure for both sites; bulk user provisioning via scripted dsadd commands; department-based security groups.                                                                                      |
| File & Print Services  | Per-user home drives, department shared drives, and a company-wide public drive — each with NTFS and share-level permissions configured and validated, with drive mapping automated via Group Policy.                         |
| Security Configuration | Domain-wide password and account lockout policy, User Account Control (UAC) hardening, and interactive logon security settings enforced through Group Policy Objects (GPOs).                                                  |
| Cloud Strategy         | A hybrid-cloud migration proposal integrating the on-premises Active Directory environment with Microsoft Azure and Microsoft 365, including Azure AD Connect synchronisation and Intune-based mobile device management.      |

Outcome

All planned infrastructure components were implemented and verified end-to-end: a fully subnetted dual-site address plan; a redundant, loop-free switching topology; a working Active Directory domain with department-aligned OUs, users, and groups; centrally managed file storage with correctly scoped permissions; an enforced security baseline; and a documented, technically justified pathway to Azure hybrid cloud adoption. The following sections present this work as a structured evidence log, followed by a summary of the competencies demonstrated.

02 NETWORK DESIGN & PLANNING

IP Addressing Strategy

Each site was allocated its own /24 supernet, split by function into a wired LAN block and a separate wireless block, so that traffic from wired workstations and Wi-Fi clients could be filtered, monitored and secured independently at the firewall. Within each block, Variable Length Subnet Masking (VLSM) was applied so that every department received a subnet sized to its actual host count rather than a single fixed mask across the board — this avoids the address wastage that a flat /24-per-department scheme would cause, while leaving headroom for departmental growth.

Design rule applied throughout: each subnet mask was chosen as the smallest block that comfortably covers the department's current headcount plus a growth margin, so that departments with similar sizes (e.g. Servers, Quantum Computing, Smart Devices — all requiring 30 usable addresses) share an identical /27 mask, while larger departments such as 3D Printing and Artificial Intelligence were given a /26 to accommodate 42+ hosts.

### Johannesburg — LAN (192.168.10.0/24)

| Department             | Subnet ID      | Usable Range | Broadcast      | Mask                  | Usable IPs |
|------------------------|----------------|--------------|----------------|-----------------------|------------|
| Servers (20)           | 192.168.10.0   | .1 – .30     | 192.168.10.31  | /27 (255.255.255.224) | 30         |
| Quantum Computing (30) | 192.168.10.32  | .33 – .62    | 192.168.10.63  | /27 (255.255.255.224) | 30         |
| 3D Printing (42)       | 192.168.10.64  | .65 – .126   | 192.168.10.127 | /26 (255.255.255.192) | 62         |
| Machine Learning (12)  | 192.168.10.128 | .129 – .142  | 192.168.10.143 | /28 (255.255.255.240) | 14         |
| Smart Devices (22)     | 192.168.10.144 | .145 – .174  | 192.168.10.175 | /27 (255.255.255.224) | 30         |

### Johannesburg — Wireless (192.168.20.0/24)

| Department      | Subnet ID     | Usable Range | Broadcast     | Mask                  | Usable IPs |
|-----------------|---------------|--------------|---------------|-----------------------|------------|
| ICTD (9)        | 192.168.20.0  | .1 – .14     | 192.168.20.15 | /28 (255.255.255.240) | 14         |
| Management (12) | 192.168.20.16 | .17 – .30    | 192.168.20.31 | /28 (255.255.255.240) | 14         |
| Operations (16) | 192.168.20.32 | .33 – .62    | 192.168.20.63 | /27 (255.255.255.224) | 30         |

### Pretoria — LAN (192.168.30.0/24)

| Department                   | Subnet ID      | Usable Range | Broadcast      | Mask                  | Usable IPs |
|------------------------------|----------------|--------------|----------------|-----------------------|------------|
| Servers (20)                 | 192.168.30.0   | .1 – .30     | 192.168.30.31  | /27 (255.255.255.224) | 30         |
| Software Development (25)    | 192.168.30.32  | .33 – .62    | 192.168.30.63  | /27 (255.255.255.224) | 30         |
| Artificial Intelligence (42) | 192.168.30.64  | .65 – .126   | 192.168.30.127 | /26 (255.255.255.192) | 62         |
| Mobile Devices (50)          | 192.168.30.128 | .129 – .190  | 192.168.30.191 | /26 (255.255.255.192) | 62         |
| Robotics (30)                | 192.168.30.192 | .193 – .222  | 192.168.30.223 | /27 (255.255.255.224) | 30         |

### Pretoria — Wireless (192.168.40.0/24)

| Department      | Subnet ID     | Usable Range | Broadcast     | Mask                  | Usable IPs |
|-----------------|---------------|--------------|---------------|-----------------------|------------|
| ICTD (10)       | 192.168.40.0  | .1 – .14     | 192.168.40.15 | /28 (255.255.255.240) | 14         |
| Management (6)  | 192.168.40.16 | .17 – .22    | 192.168.40.23 | /29 (255.255.255.248) | 6          |
| Operations (16) | 192.168.40.24 | .25 – .54    | 192.168.40.55 | /27 (255.255.255.224) | 30         |

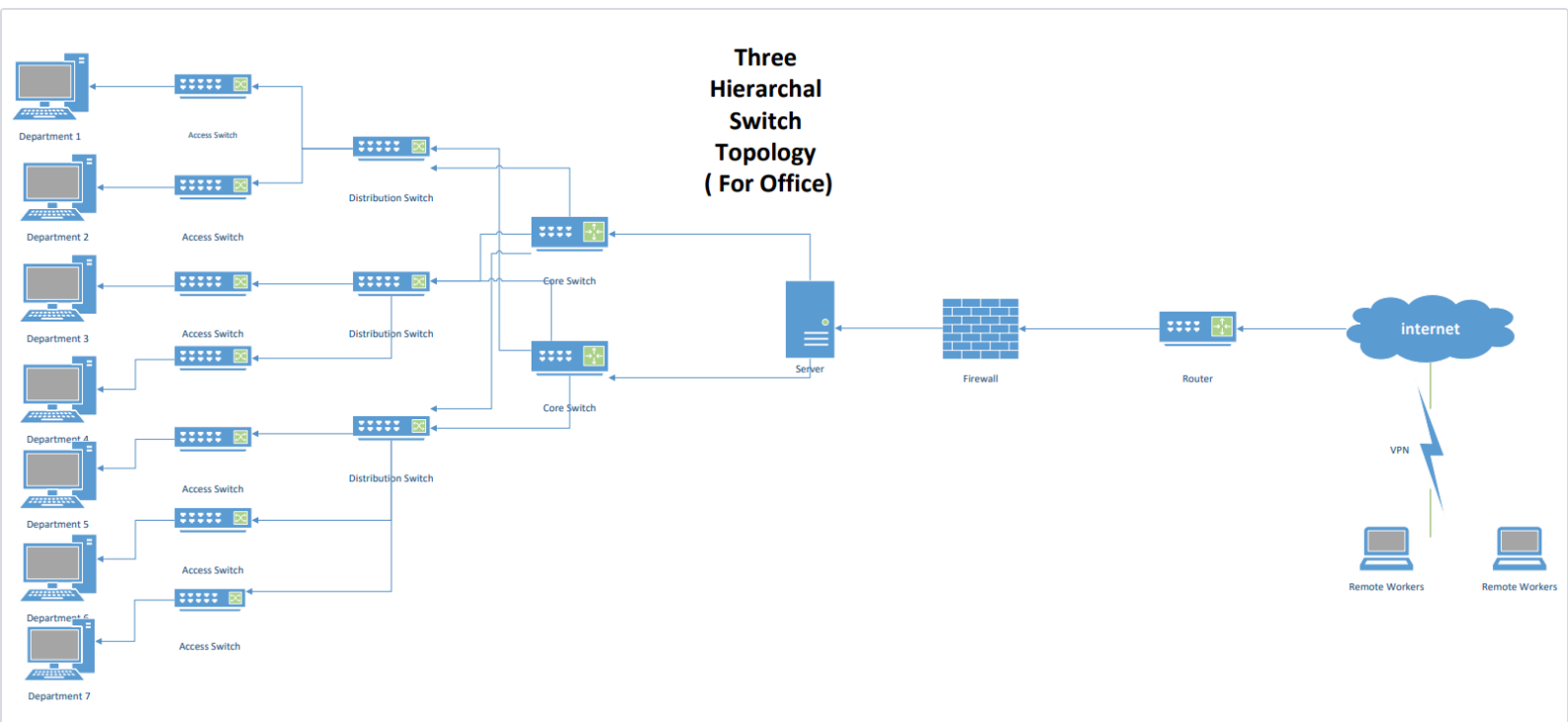
### Hierarchical Switch Topology & Loop Prevention

Each office was designed around a three-tier hierarchical switching model — access switches at the department/desk level, aggregated upward through distribution switches, and consolidated at a redundant core switch pair connected to the site server, firewall and edge router. This tiered structure keeps broadcast domains contained at the access layer, provides clear points for applying department-level ACLs and VLANs, and gives each site a predictable upgrade path as department headcount grows. The two offices are connected through their respective firewalls and an SDN switch, with VPN access provisioned for remote workers at both sites.

Because the topology intentionally includes redundant links between distribution and core switches (for resilience against a single link or switch failure), the design is exposed to Layer 2 bridging loops and the broadcast storms they cause. Spanning Tree Protocol (STP) was applied as the mitigating control: STP elects a single root bridge for the switched network, calculates the lowest-cost path from every other switch to that root, and then blocks the redundant, higher-cost ports that would otherwise form a loop — while keeping them on standby to fail over automatically if the primary path goes down. This preserves the resilience benefit of the redundant links without the broadcast-storm risk, and is the same underlying mechanism (extended by Rapid STP / Multiple STP in modern deployments) used in production enterprise switching designs.

**Figure 2.1: Per-Office Hierarchical Switch Topology**

Three-tier design for a single office: department-level access switches feed into distribution switches, which converge on a redundant core switch pair connected to the local server, firewall, and edge router — the point at which STP root-bridge election and redundant-path blocking is applied.

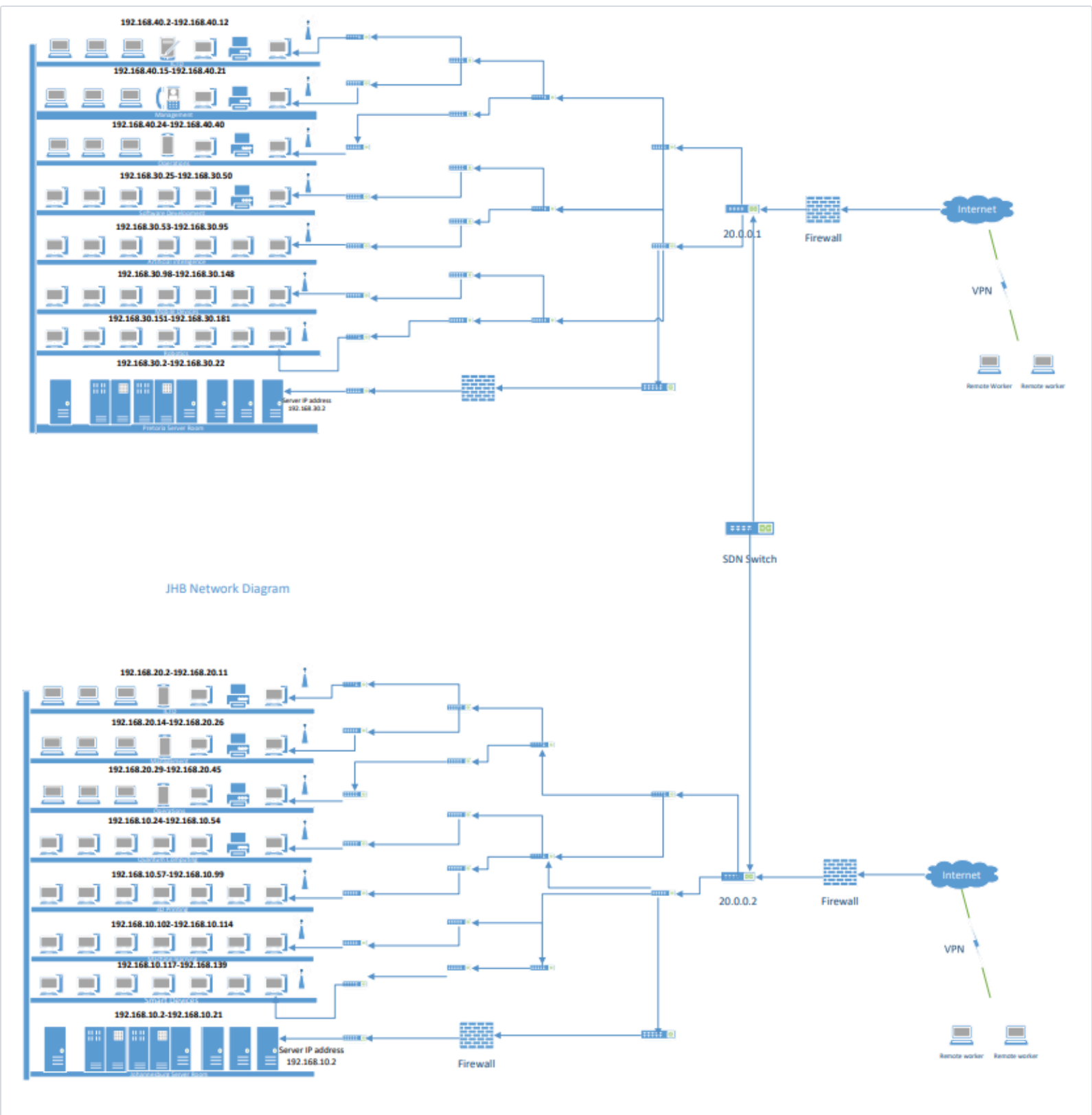


**Figure 2.2: Dual-Site Network Diagram — Johannesburg & Pretoria Interconnect**

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Full logical diagram showing both offices' department subnets (annotated with their allocated IP ranges from the tables above), each site's firewall and server, and the SDN-switched link that joins the two sites — with VPN-connected remote workers shown entering through each site's firewall.



### 03

## EVIDENCE LOG — ACTIVE DIRECTORY IMPLEMENTATION

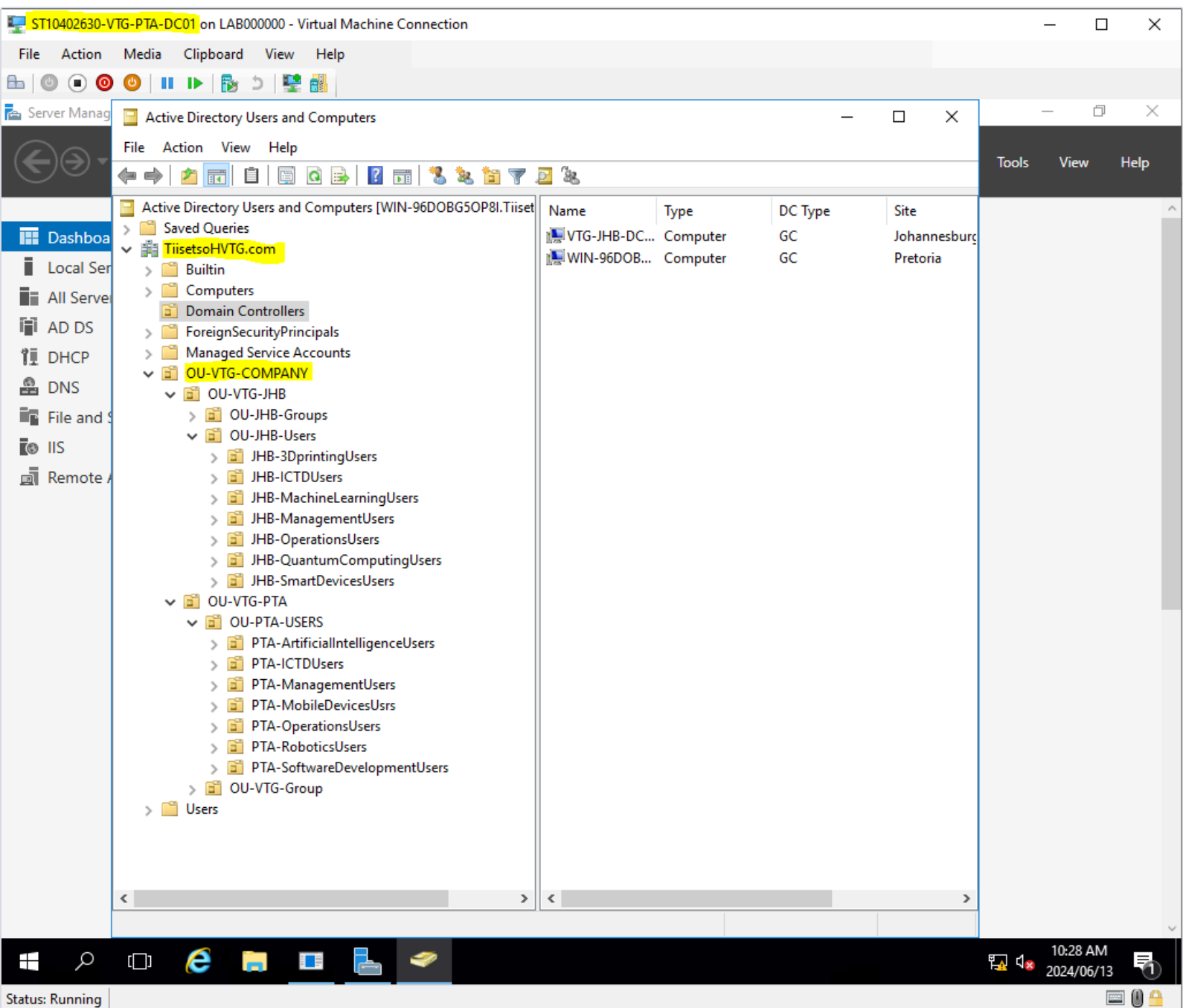
This section documents the build-out of the Active Directory domain structure on the Johannesburg and Pretoria domain controllers, covering Organisational Unit (OU) design, bulk user provisioning, and security group creation.

### Organisational Unit Design

The OU hierarchy was structured to mirror the physical and departmental design established in the network planning phase: a top-level company OU containing separate site OUs for Johannesburg (OU-VTG-JHB) and Pretoria (OU-VTG-PTA), each further divided into a department-aligned Users OU per team — for example JHB-3DprintingUsers, JHB-MachineLearningUsers, PTA-ArtificialIntelligenceUsers, and PTA-RoboticsUsers — matching the subnets defined for each department in Section 02. This structure keeps Group Policy scoping, delegation, and account administration aligned to department boundaries, which is what allows department-specific policies (drive mappings, security settings) to be targeted precisely without affecting unrelated teams.

**Figure 3.1: OU Structure Implemented in Active Directory Users and Computers**

Active Directory Users and Computers console on the Pretoria domain controller (ST10402630-VTG-PTA-DC01), showing the completed OU-VTG-COMPANY hierarchy with both site branches (OU-VTG-JHB, OU-VTG-PTA) expanded to reveal their department-level user OUs. Both domain controllers are visible under Domain Controllers, registered to their respective AD sites (Johannesburg, Pretoria), confirming the multi-site domain topology is active.



## Bulk User Provisioning

Rather than creating accounts individually through the GUI — impractical at the scale required for each department — user accounts were provisioned using scripted `dsadd` user commands run from an elevated command prompt, with each department's user list prepared as a batch file. Each command specifies the full distinguished name (CN/OU/DC path) placing the account directly into its correct department OU, along with the first name, logon name (`-fn`, `-ln`), User Principal Name (`-upn`), and an initial password with a just-then-power-group-over-forced password change at next logon. This approach is 95% faster and

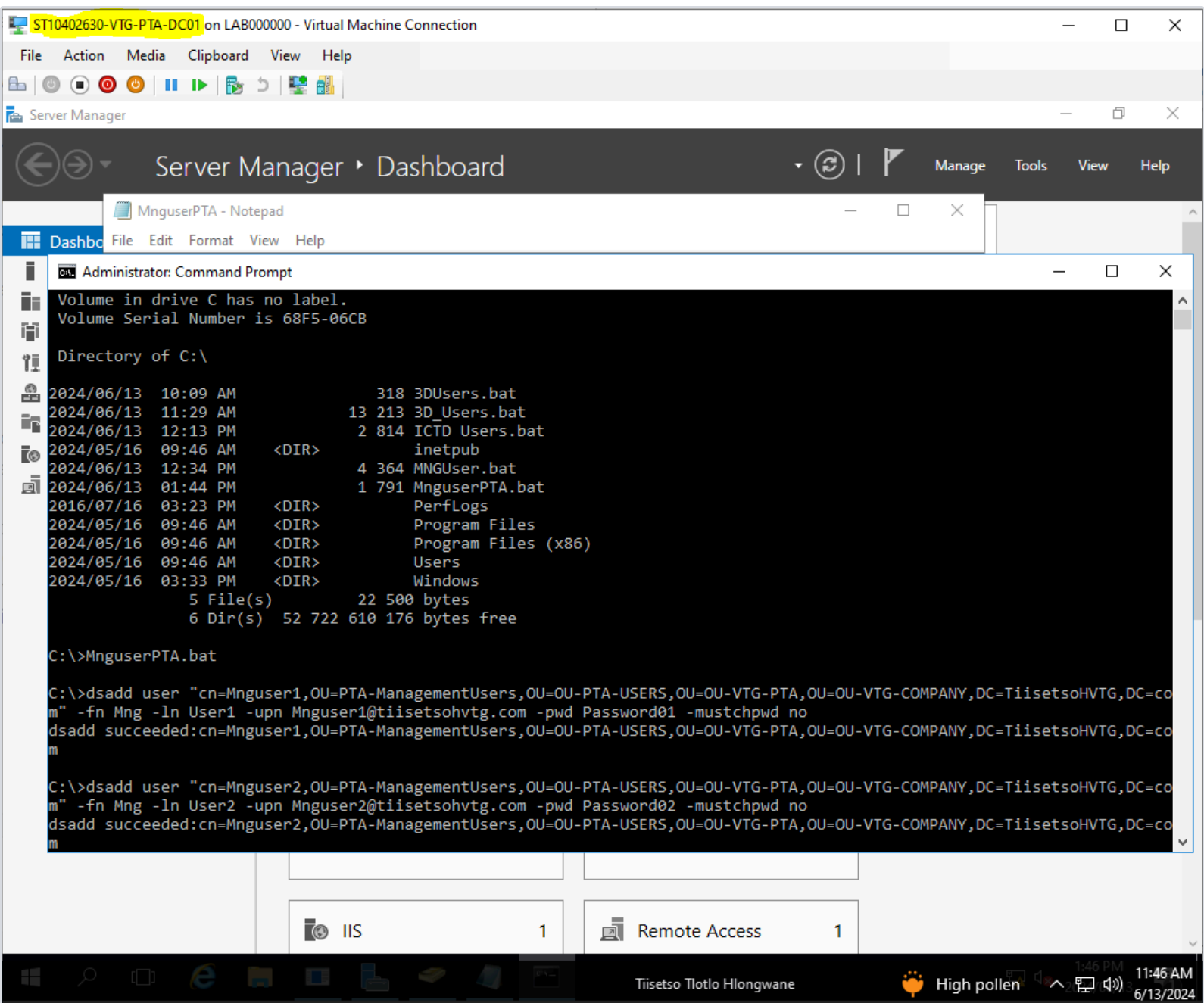


**Figure 3.2: Batch User-Creation Script**

Notepad view of the prepared dsadd user batch script for the Johannesburg 3D Printing department, showing each user's distinguished name targeting OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY within the TiisetsoHVTG.com domain, with a shared naming convention (3DUser1–3DUser36) applied for consistency and easy identification during audits.

```
dsadd user "cn=3DUser2,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser3,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser4,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser5,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser6,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser7,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser8,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser9,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser10,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser11,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser12,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser13,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser14,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser15,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser16,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser17,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser18,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser19,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser20,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser21,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser22,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser23,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser24,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser25,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser26,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser27,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser28,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser29,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser30,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser31,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser32,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser33,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser34,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser35,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser36,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
dsadd user "cn=3DUser37,OU=JHB-3DprintingUsers,OU=OU-JHB-Users,OU=OU-VTG-JHB,OU=OU-VTG-COMPANY,DC=TiisetsoHVTG,DC=com" -fn 3D
```

**Figure 3.3: Script Execution — Users Committed to the Domain**  
Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence  
Command prompt output on the Pretoria DC confirming successful execution of the batch scripts (3DUsers.bat, ICTD Users.bat, MngUserPTA.bat). Each dsadd succeeded line confirms the corresponding account was created and correctly placed into its target OU, with UPN suffixes generated automatically (e.g. Mnguser1@tiisetsohvtg.com).



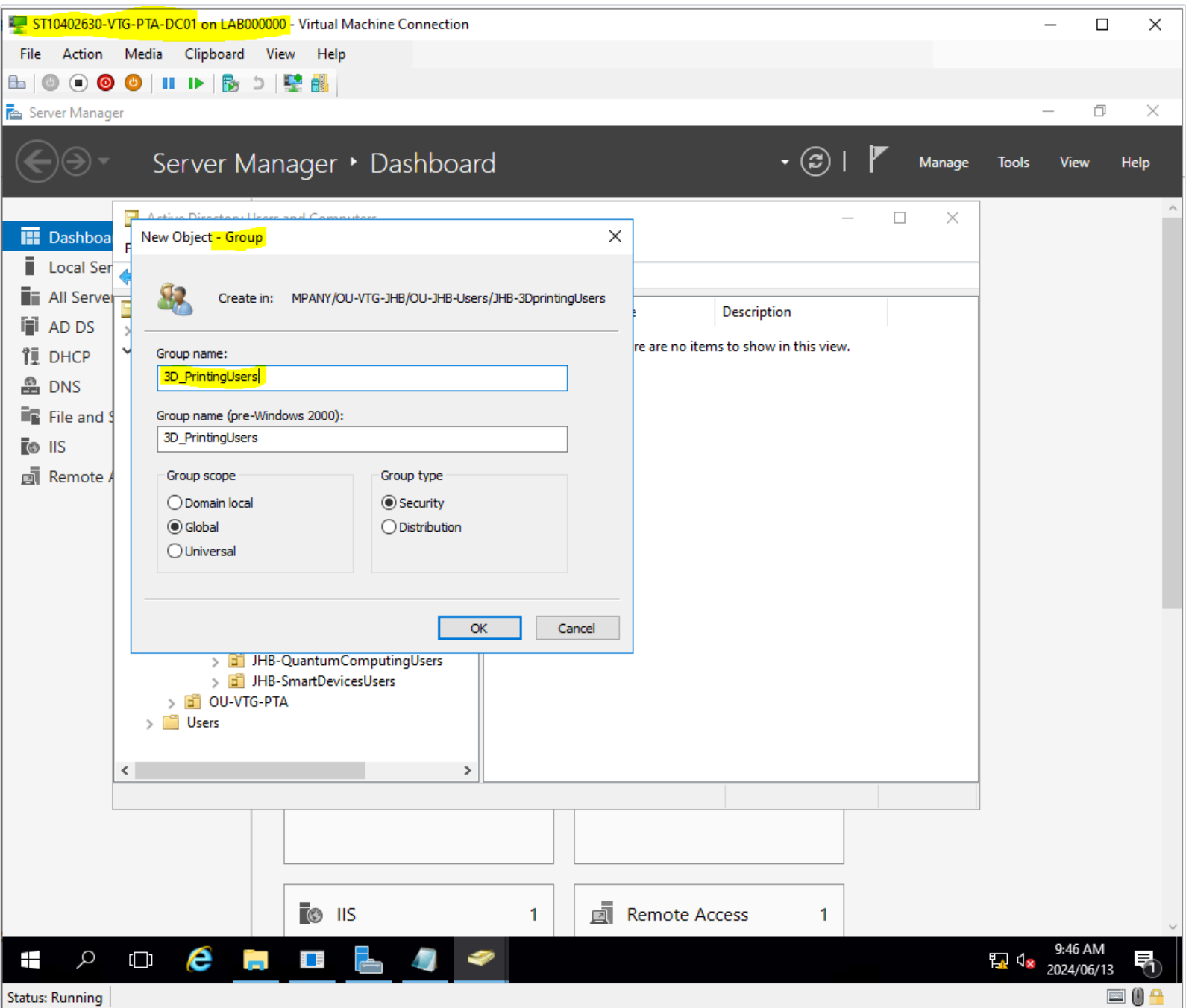
## Security Groups

A department-scoped Global Security Group was created for each team (e.g. 3D\_PrintingUsers) and populated with the corresponding user accounts. Security groups — rather than applying permissions to individual users — are the mechanism used throughout the rest of this project (file share permissions in Section 04, and GPO security filtering) so that access control scales with department membership changes instead of requiring per-user maintenance.



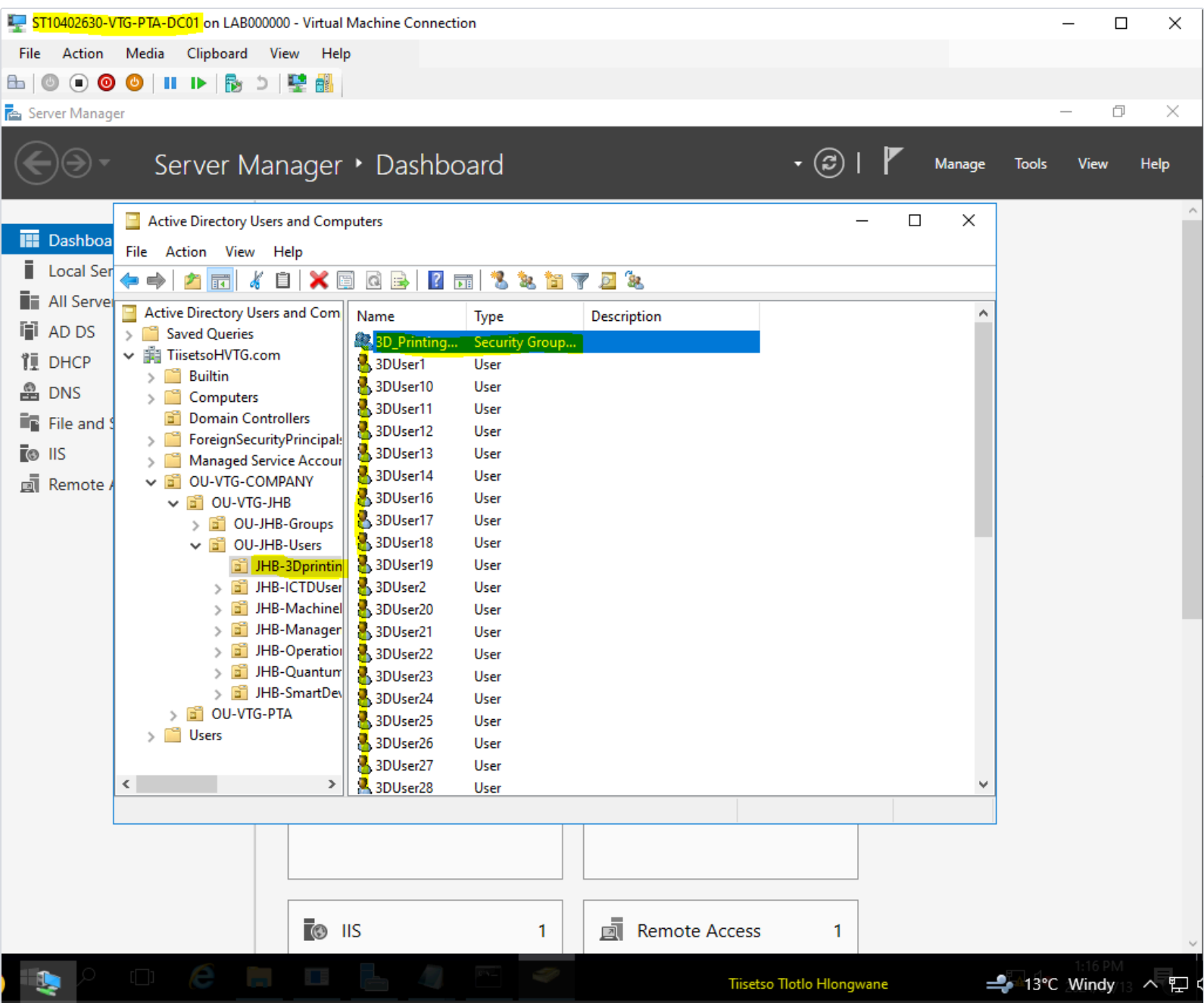
**Figure 3.4: Security Group Creation**

**Enterprise Network Infrastructure & Active Directory Deployment** — Portfolio of Evidence  
New Object – Group dialog showing the 3D\_PrintingUsers group being created within the JHB-3dprintingUsers OU, with Group scope set to **Global** and Group type set to **Security** — the standard combination for a department access-control group within a single-domain environment.



**Figure 3.5: Group Membership Verified** — Portfolio of Evidence  
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Active Directory Users and Computers view of the JHB-3Dprinting OU confirming the 3D\_Printing Security Group and its full membership of provisioned user accounts (3DUser1–3DUser28 shown) are correctly listed and available for use in downstream permission assignment.



## 04 EVIDENCE LOG — FILE & PRINT SERVICES

This section covers the storage layer built on top of the Active Directory structure from Section 03: per-user home drives, a department shared drive, and a company-wide public drive — each provisioned with appropriate NTFS and share permissions and, where applicable, deployed automatically via Group Policy drive mapping.

### 4.1 — Home Drive Provisioning

A dedicated NTFS volume was created on the domain controller to host user storage, formatted and labelled as a distinct data disk separate from the system volume. A **Home Folders** share was configured on this volume with Domain Users granted Full Control at the share level, and each user's Active Directory profile was configured with a Home Folder connection pointing to `\\...\Home_Folder$\%username%` — using the `%username%` environment variable so that Windows automatically creates and maps each user to their own private subfolder at logon, without needing a manually created folder per account.

**Figures 4.1 – 4.2: Storage Volume Created and Confirmed Visible** Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence Tietseto Tlotlo Hlongwane

Disk Management confirms the new simple volume (10GB, NTFS, labelled tiisetsohvtg.com-storage) was created successfully on the second disk attached to the domain controller; File Explorer then confirms the volume is mounted and visible as the S: drive, ready to host the Home Folders and department shares built in the steps below.

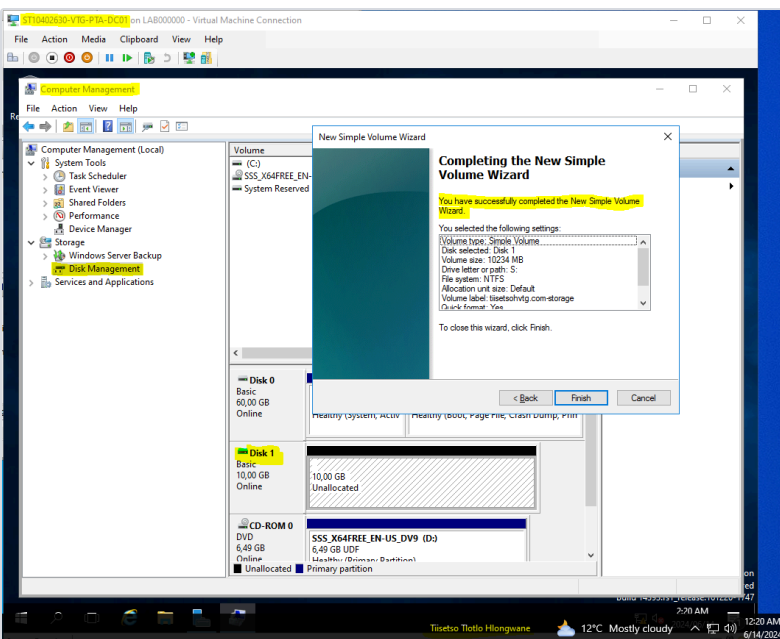


Fig. 4.1

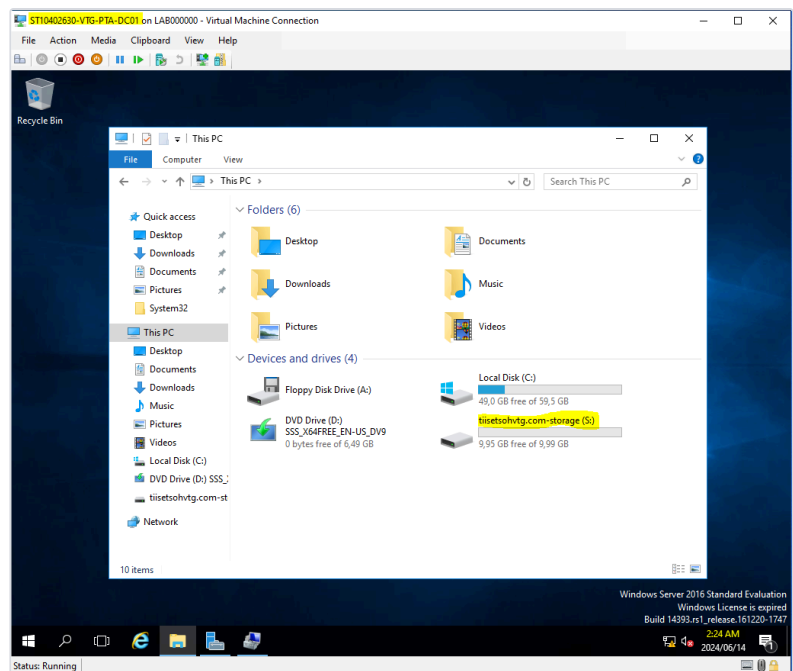


Fig. 4.2

### Figures 4.3 – 4.4: Share Permissions and Per-User Home Folder Path Configured

Share Permissions on the Home\_Folder\$ share are set to grant the TIISSETSOHVTG\Domain Users group Full Control, Change, and Read at the share level; the corresponding user profile configuration (applied across the department's users via Properties for Multiple Items) sets the Home Folder connection to drive S: mapped to \\...\Home\_Folder\$\%username%, so each user is automatically assigned a private, isolated folder inside the shared volume at next logon.

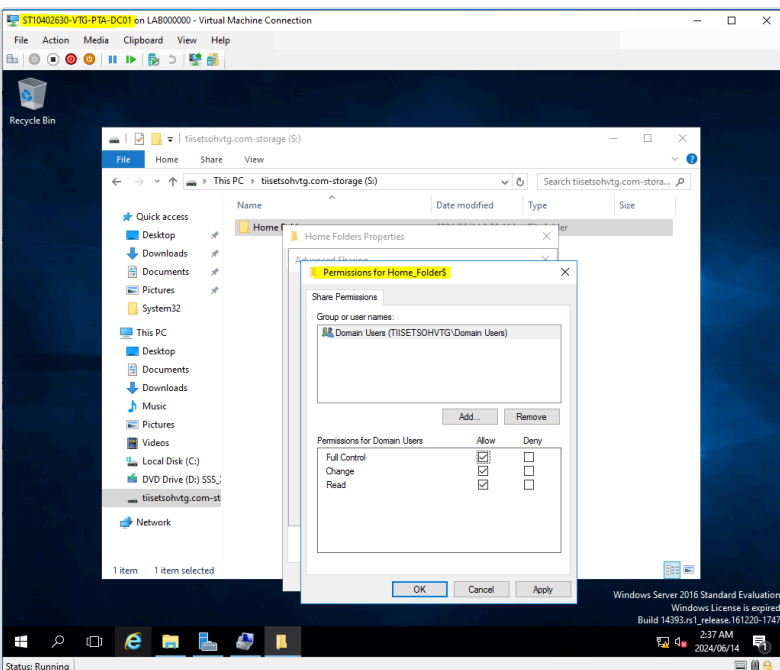


Fig. 4.3

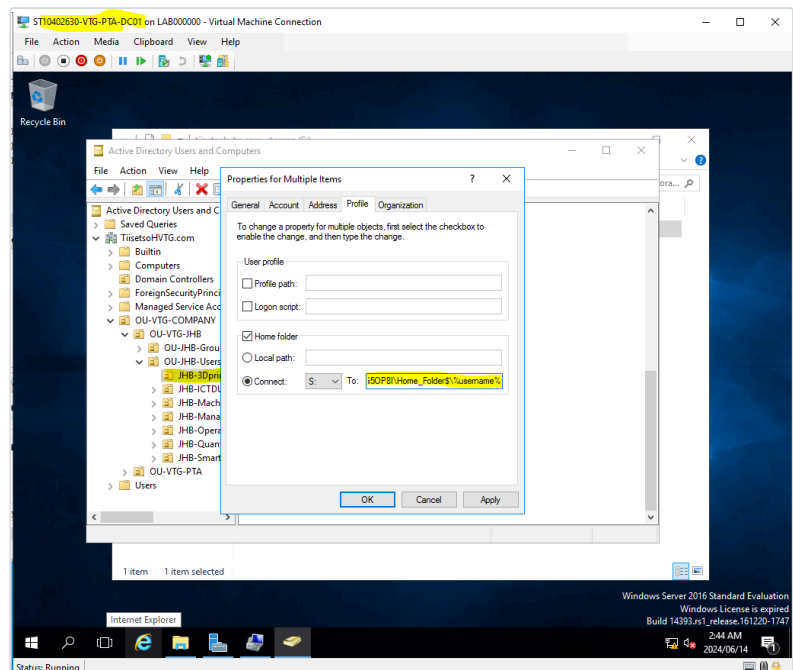


Fig. 4.4

**Figures 4.5 – 4.6: Home Folders Populated and Verified from a Client Machine** Tietseto Tlotlo Hlongwane  
**Enterprise Network Infrastructure & Active Directory Deployment** — Portfolio of Evidence  
The Home Folders directory on the server confirms individual folders were auto-provisioned for each user account at first logon; from a domain-joined client workstation (ST10402630-VTG-PTA-CLT01), the mapped drive is confirmed visible under Network Locations as both the public drive and the individual user's home share — validating that the server-side configuration correctly resolves to a working client-side mapping.

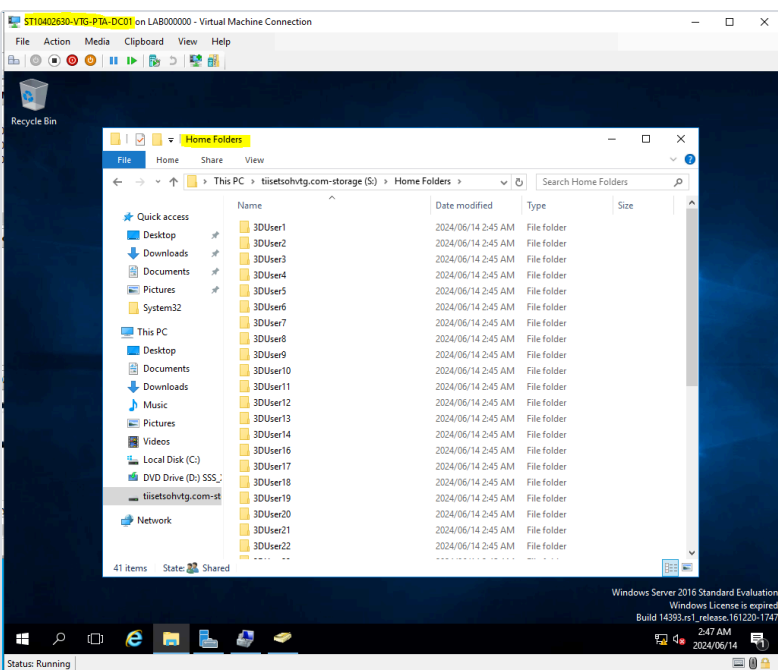


Fig. 4.5

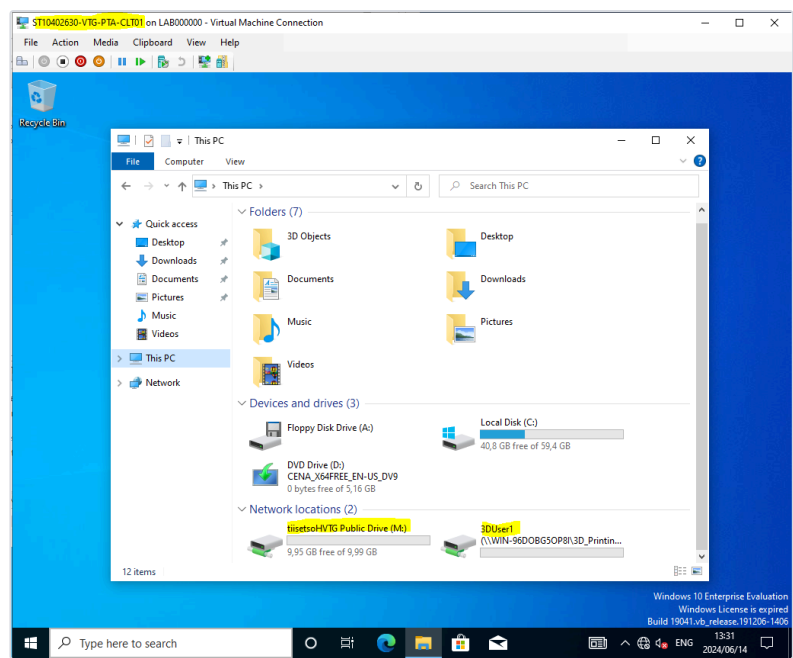


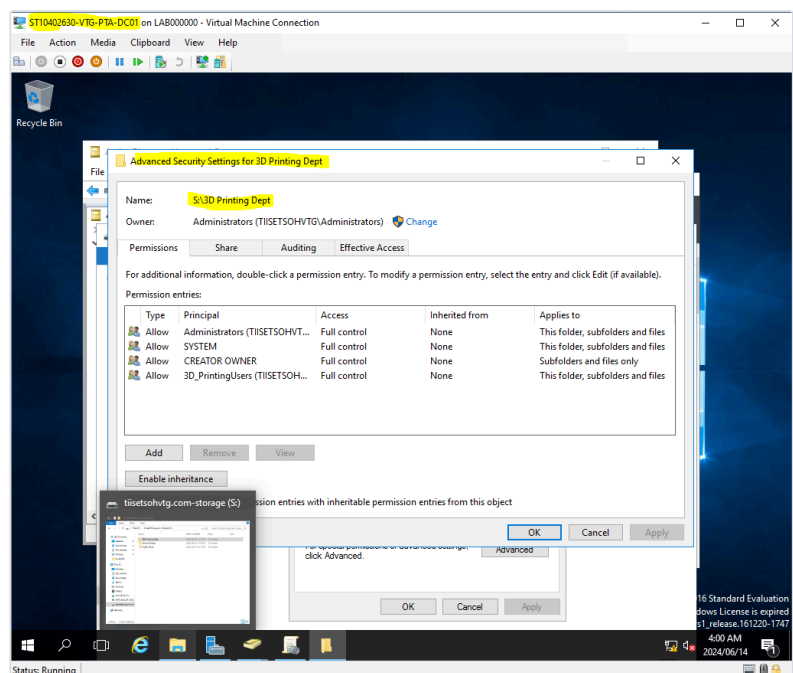
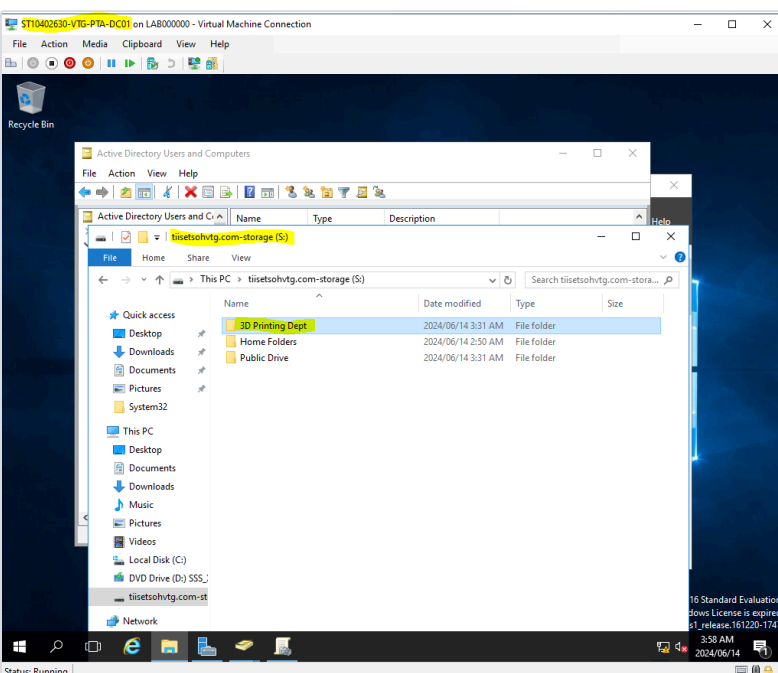
Fig. 4.6

## 4.2 — Department Shared Drive

In addition to private home folders, a department-level shared drive (**3D Printing Dept**) was created so that team members can collaborate on shared files without exposing them to the wider company. NTFS permissions on the folder were configured to grant the **3D\_PrintingUsers** security group (created in Section 03) Full Control, to that folder, its subfolders, and files — while share permissions mirror the same access for the same group. Using the security group rather than individual users means membership changes in Active Directory automatically flow through to folder access with no additional administration.

### Figures 4.7 – 4.8: Department Drive Created and NTFS Permissions Applied

The **3D Printing Dept** folder is created inside the storage volume; Advanced Security Settings confirm NTFS permission entries for Administrators, SYSTEM, CREATOR OWNER, and the **3D\_PrintingUsers** group — the last of which grants the department Full Control over the folder, its subfolders, and files.



**Figures 4.9 – 4.10: Share-Level Permissions and Active Directory Deployment** — Portfolio of Evidence

Share Permissions for 3D\_Printing\_Dept\$ confirm Full Control, Change, and Read are granted to the 3D\_PrintingUsers group and Administrators; the department drive is then confirmed populated with each user's working folder, verifying the share is correctly scoped and accessible to the department only.

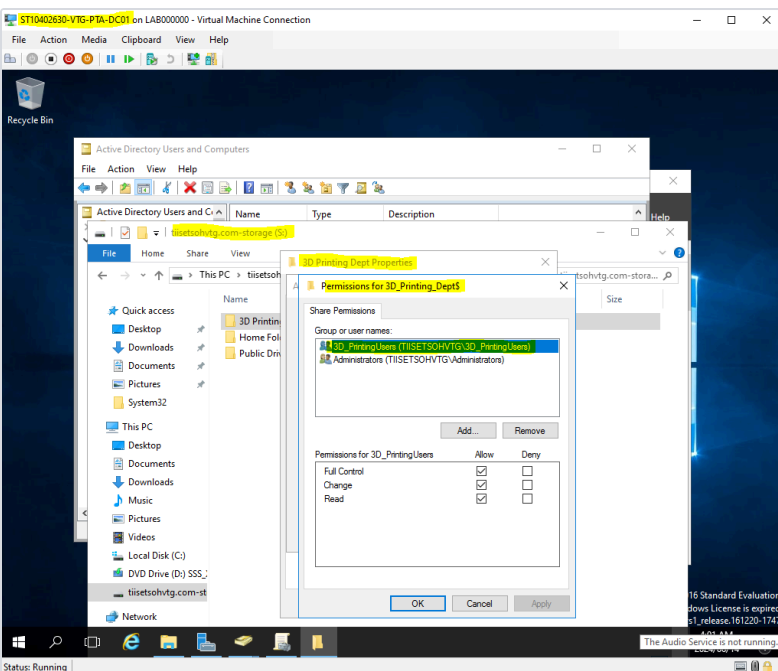


Fig. 4.9

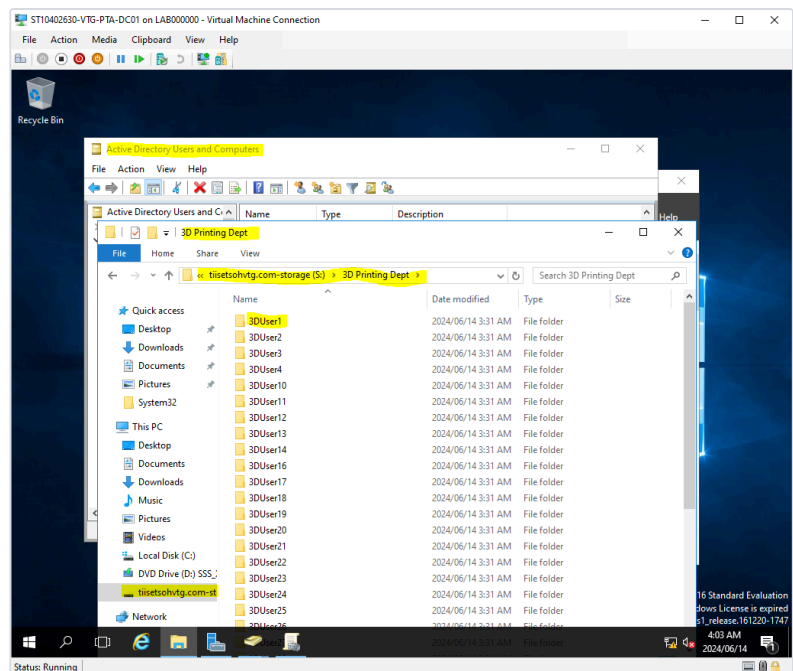


Fig. 4.10

### 4.3 — Company-Wide Public Drive

A third share — a company-wide **Public Drive** — was created for resources that need to be accessible to all staff regardless of department, with permissions granted to the broader Domain Users group rather than a single department group. Unlike the home and department drives, this share was deployed via a **Group Policy Object** (PublicDrive Share) using Group Policy Preferences → Drive Maps, so that the mapping is applied automatically and consistently to every domain-joined machine without requiring manual configuration on each client — the GPO was linked at the OU-VTG-COMPANY level and scoped to the Domain Users group via Security Filtering, ensuring company-wide reach.

#### Figures 4.11 – 4.12: Public Drive Created and Share Permissions Set

The Public Drive folder is added alongside Home Folders and the department drive on the storage volume; Share Permissions confirm Full Control, Change, and Read are granted to Domain Users, making the resource accessible company-wide rather than restricted to a single department.

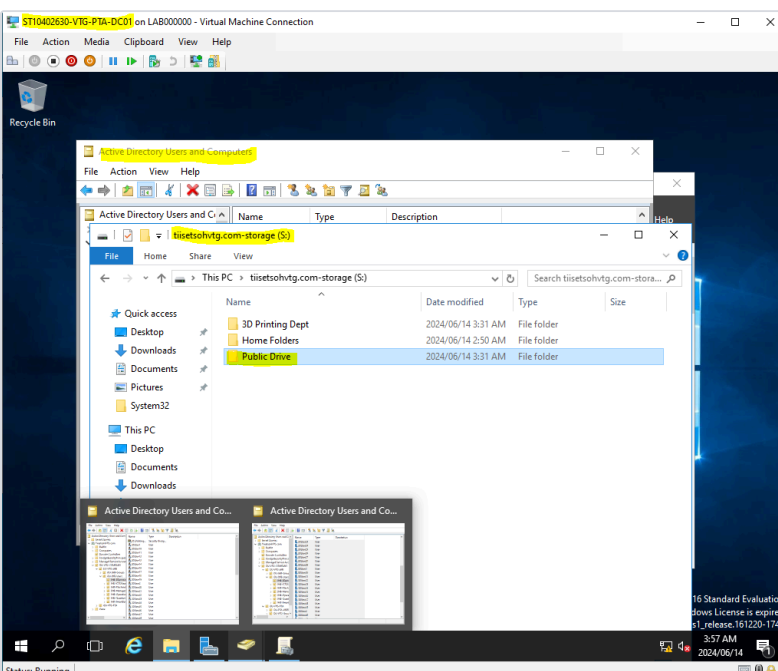


Fig. 4.11

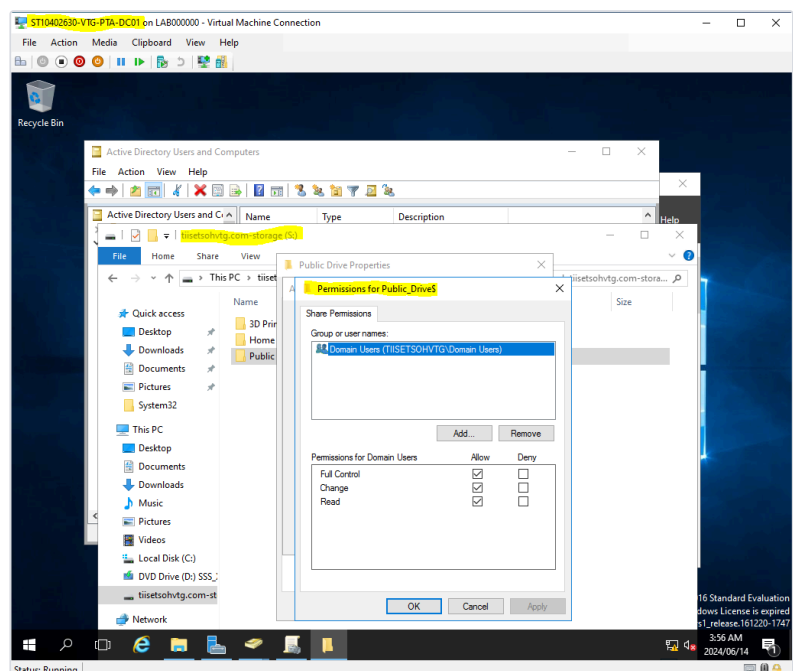


Fig. 4.12



**Figures 4.13 – 4.14: Group Policy Drive Mapping Configured and Linked** Tietso Tlotlo Hlongwane  
**Enterprise Network Infrastructure & Active Directory Deployment** — Portfolio of Evidence  
 Group Policy Management Editor showing a new Drive Map preference targeting \\WIN-96D0BG50P81\\Public\_Drive\$, set to map as drive M: with the label *tietsoHVTG Public Drive* and Reconnect enabled so the mapping persists across logoffs; the PublicDrive Share GPO is then confirmed linked and enforced at the OU-VTG-COMPANY level, with Security Filtering scoped to the Domain Users group.

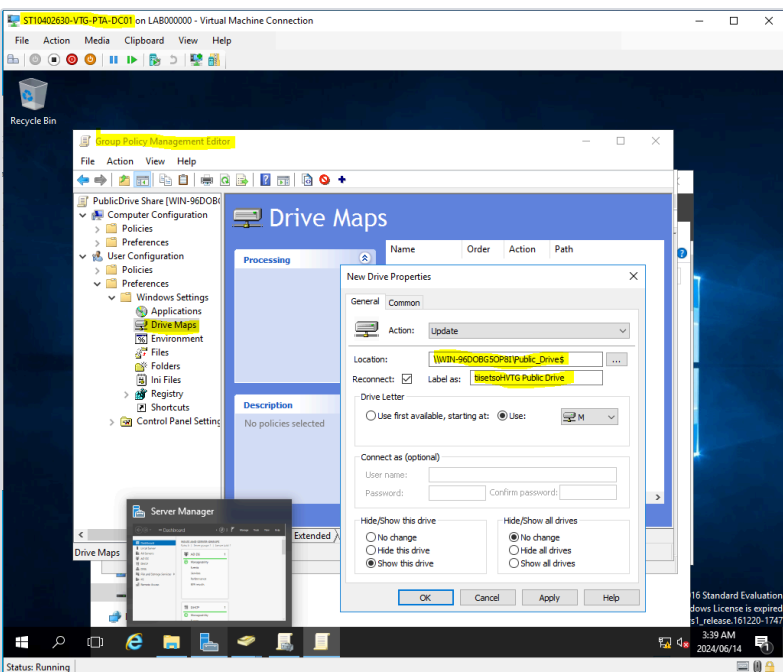


Fig. 4.13

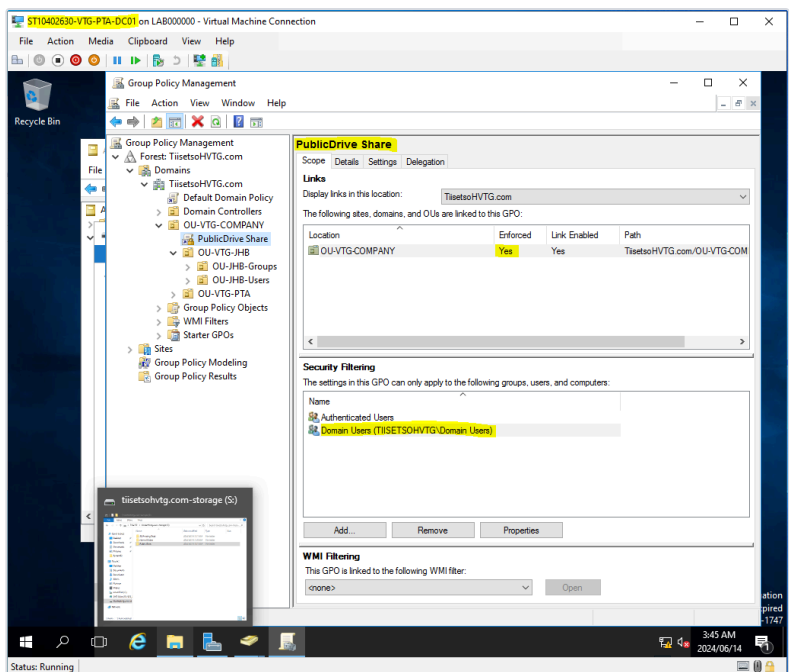


Fig. 4.14

**Figures 4.15 – 4.16: Policy Applied and Confirmed on a Client Workstation**  
 gpupdate /force is run on the domain controller to push the new policy immediately rather than waiting for the default refresh interval; on a client workstation, the mapped **tietsoHVTG Public Drive (M:)** is then confirmed visible under Network Locations — proving the GPO-based drive mapping was delivered and applied correctly without any manual client-side configuration.

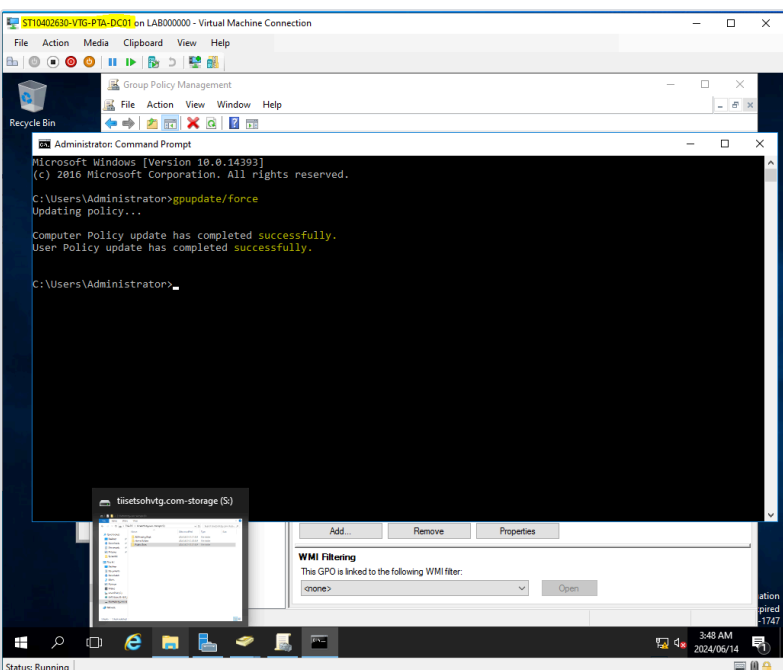


Fig. 4.15

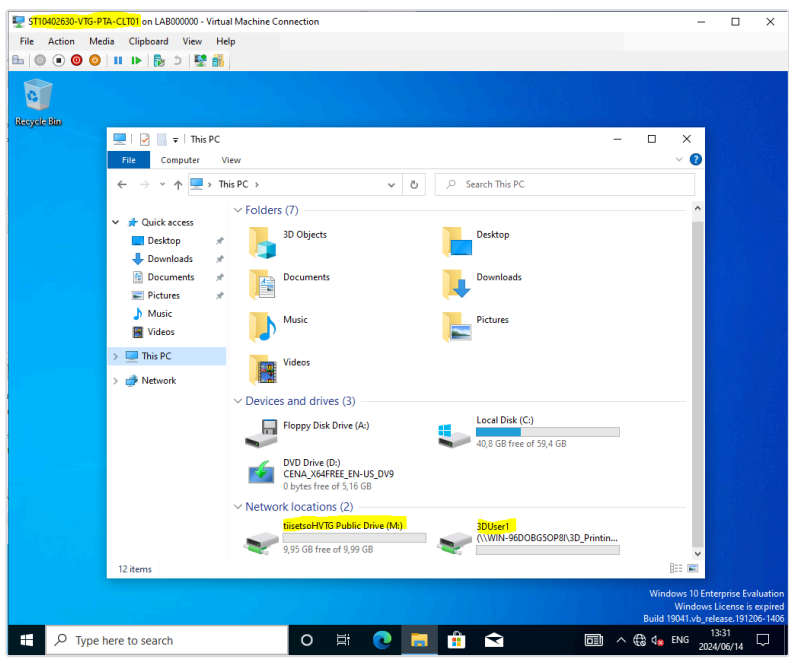


Fig. 4.16

## 4.4 — Consolidated NTFS & Share Permission Review

As a final validation step, the effective permissions across the Home Folders structure were reviewed together at the NTFS level, confirming that permission inheritance and explicit group assignments were both correctly applied and did not conflict.



**Figures 4.17 – 4.18: Home Folder NTFS Permissions Reviewed** — Portfolio of Evidence Tiisetso Tlotlo Hlongwane

Enterprise Network Infrastructure & Active Directory Deployment — Permission Entry dialog confirms Domain Users is granted Full Control, Modify, Read & Execute, List Folder Contents, Read, and Write on the Home Folders share; the Advanced Security Settings view confirms this entry sits alongside the default Administrators, SYSTEM, and CREATOR OWNER entries with correct inheritance applied to the folder, its subfolders, and files — completing the permission model for the storage layer.

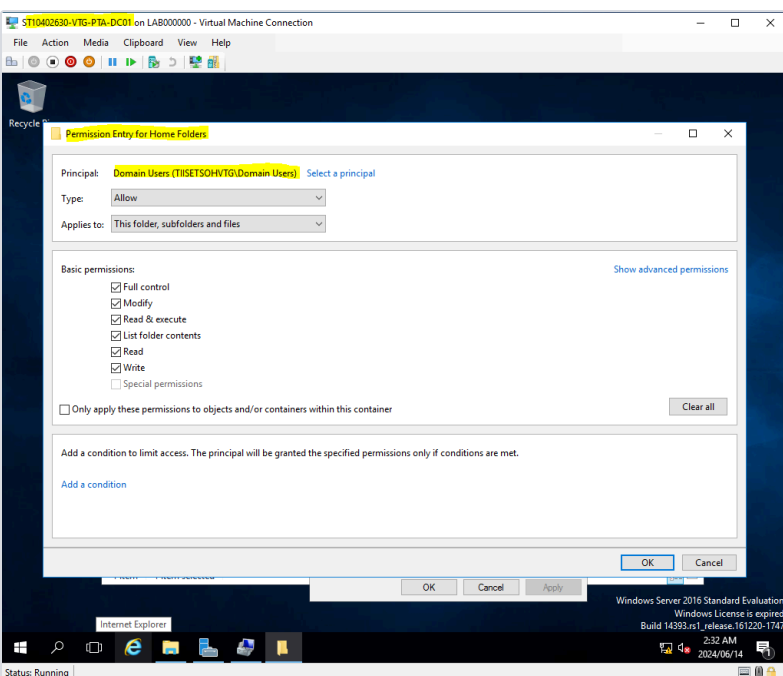


Fig. 4.17

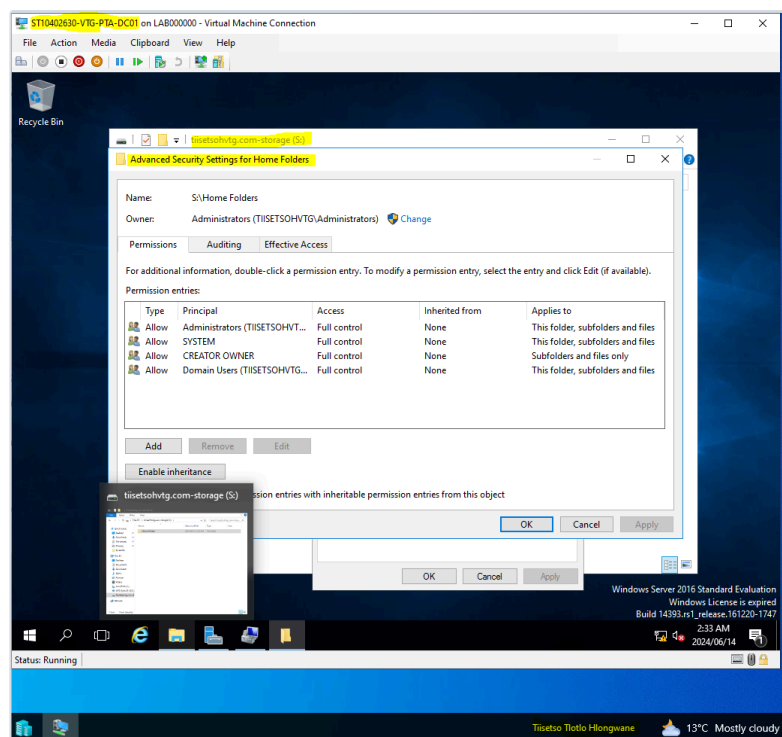


Fig. 4.18

## 05 EVIDENCE LOG — SECURITY POLICY CONFIGURATION

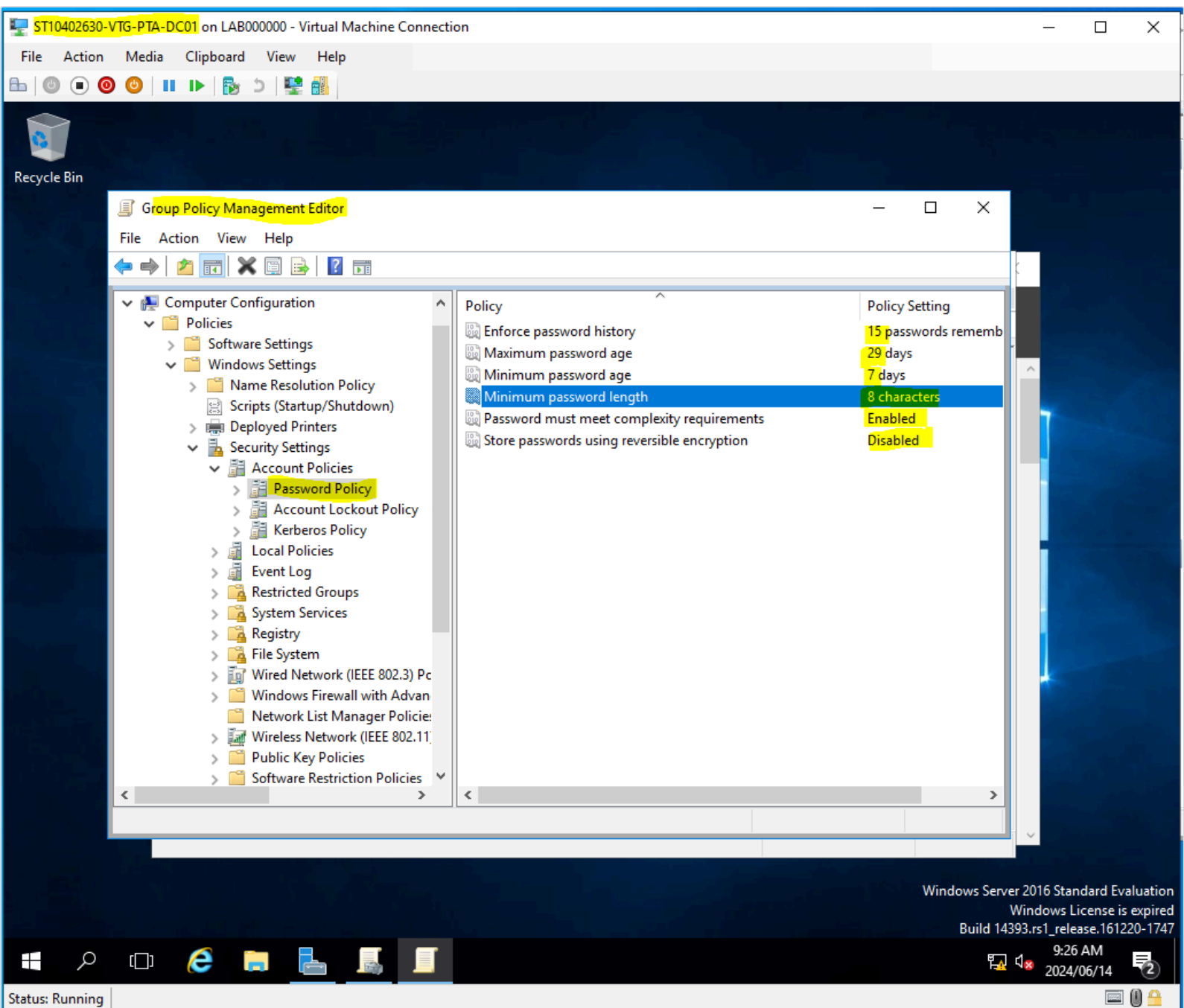
A domain-wide security baseline was enforced through Group Policy Objects linked at the Computer Configuration level, covering password strength, account lockout behaviour, User Account Control, and interactive logon protections — applied consistently across both sites rather than left to per-machine local policy.

### 5.1 — Password Policy

The domain password policy was configured to require a minimum length of 8 characters with complexity requirements enabled (a mix of character types), a password history of 15 remembered passwords to prevent immediate reuse, a maximum password age of 29 days to force periodic rotation, and a minimum password age of 7 days to prevent users from cycling rapidly back to a familiar password. Reversible encryption storage was left disabled, as there was no application dependency requiring it and enabling it would weaken password storage security for no operational benefit.

**Figure 5.1: Domain Password Policy Configured**

Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence Tisetso Tlotlo Hlongwane  
Group Policy Management Editor showing the Password Policy node under Account Policies, with Enforce Password History set to 15 passwords, Maximum Password Age to 29 days, Minimum Password Age to 7 days, Minimum Password Length to 8 characters, Complexity Requirements enabled, and Reversible Encryption disabled.

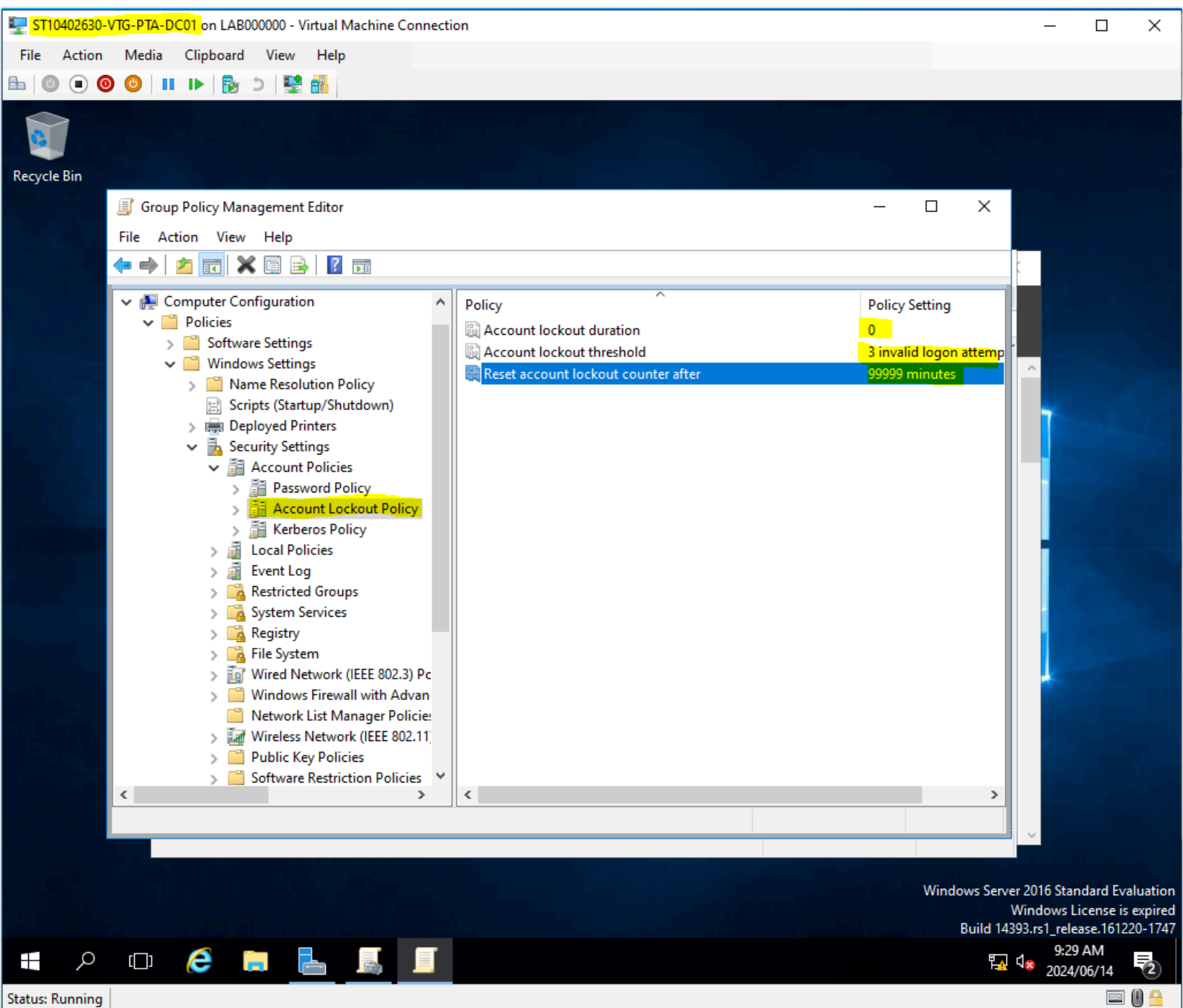


## 5.2 — Account Lockout Policy

Account lockout was configured with a threshold of 3 invalid logon attempts before an account is locked, mitigating brute-force and password-guessing attacks against domain accounts. The lockout counter reset window was set to 99999 minutes (effectively requiring administrator intervention to unlock rather than an automatic timeout), which was a deliberate choice to force explicit administrative review of repeated failed-logon events rather than allowing silent self-recovery.

**Figure 5.2: Account Lockout Policy Configured**

Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence, Tiisetso Tlotlo Hlongwane  
Account Lockout Policy node showing Account Lockout Threshold set to 3 invalid logon attempts and Reset Account Lockout Counter After set to 99999 minutes, requiring administrative unlock following repeated failed authentication attempts.



### 5.3 — User Account Control (UAC) Hardening

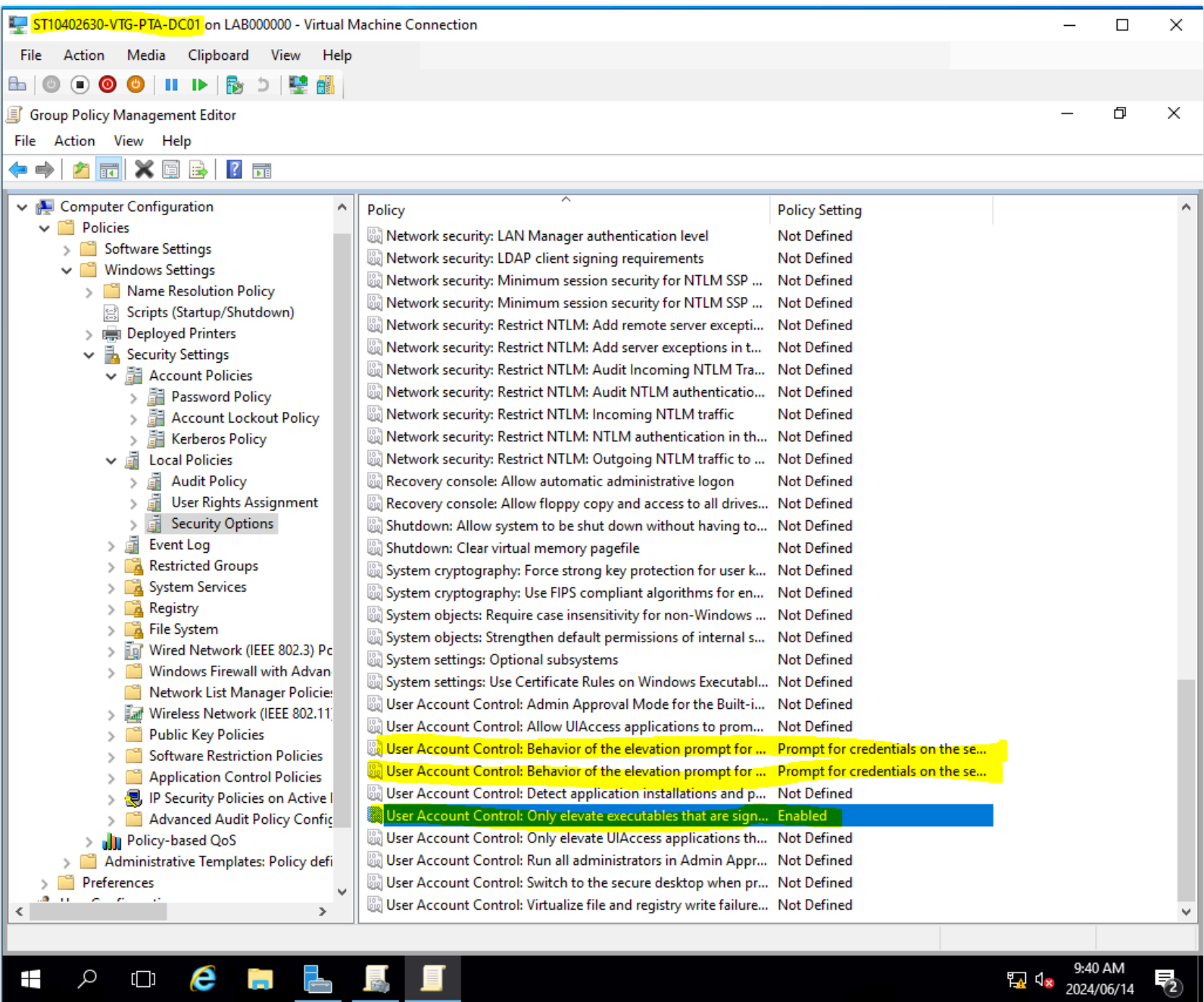
User Account Control settings under Security Options were reviewed and hardened beyond the Windows default: both the administrator and standard-user elevation prompt behaviours were set to prompt for credentials on the secure desktop, and *User Account Control: Only elevate executables that are signed and validated* was explicitly enabled — blocking unsigned or unvalidated executables from silently requesting elevation, which reduces the attack surface for privilege-escalation malware on domain-joined machines.

**Figure 5.3: UAC and Local Security Options Hardened**

Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence

Tiisetso Tlotlo Hlongwane

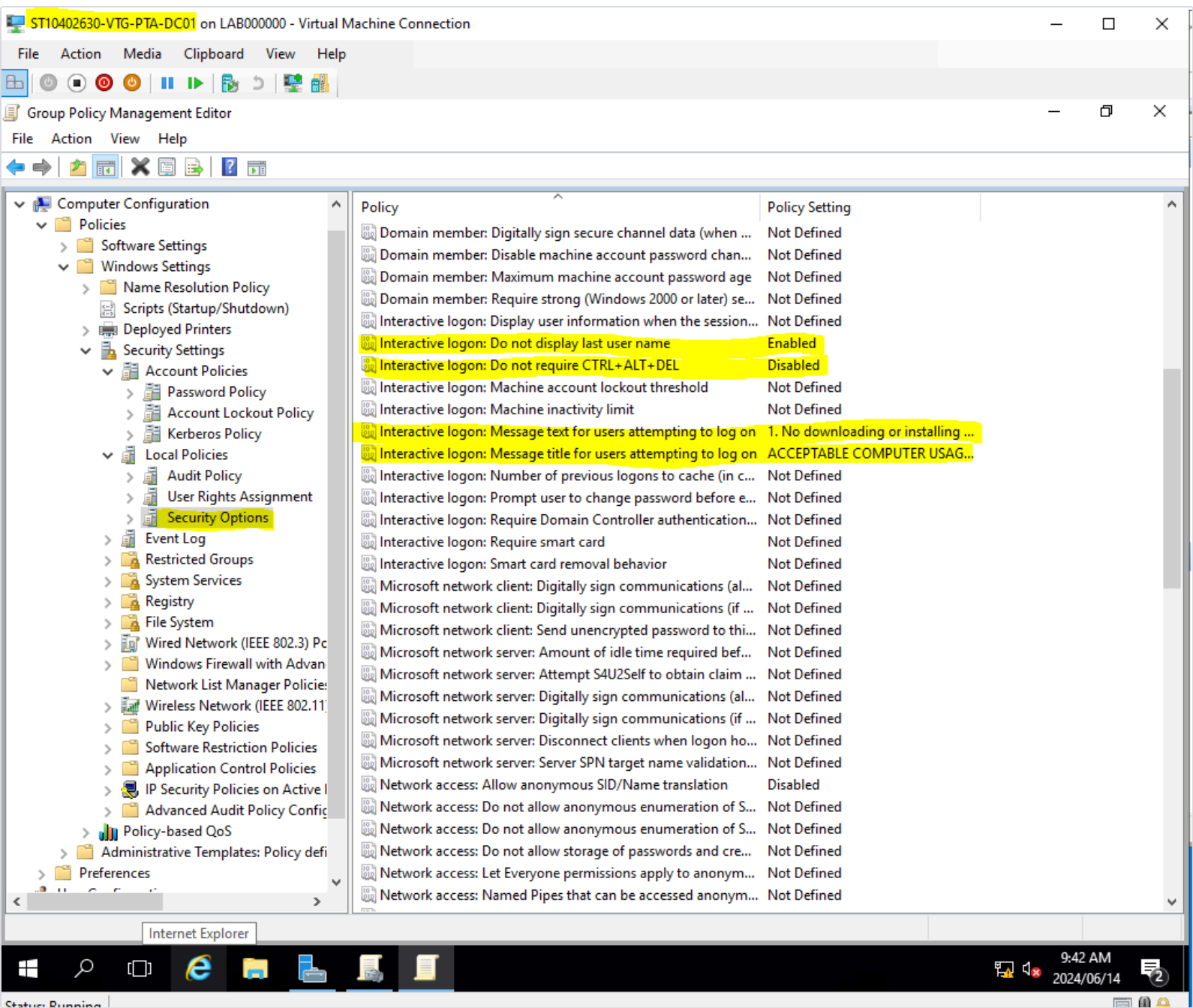
Security Options list showing User Account Control: Behavior of the elevation prompt (for both standard users and administrators in Admin Approval Mode) set to *Prompt for credentials on the secure desktop*, and Only Elevate Executables That Are Signed and Validated set to **Enabled**.



## 5.4 — Interactive Logon Security

Interactive logon settings were configured to suppress the display of the last signed-in username at the logon screen — reducing the information available to anyone attempting unauthorised access to a domain workstation — and a formal logon banner was enabled, presenting an "Acceptable Computer Usage" message and requiring explicit acknowledgement before a session begins. This combination addresses both a technical control (information disclosure at the lock screen) and a policy/compliance control (documented user acknowledgement of acceptable use).

**Figure 5.4: Interactive Logon Policy Configured** — Portfolio of Evidence  
 Enterprise Network Infrastructure & Active Directory Deployment — Portfolio of Evidence  
 Security Options showing Interactive Logon: Do Not Display Last User Name set to **Enabled**, and Message Title / Message Text for Users Attempting to Log On configured with an Acceptable Computer Usage notice, displayed to every user prior to authentication.



## 06 CLOUD STRATEGY — AZURE HYBRID INTEGRATION

Beyond the on-premises build, this portfolio includes a strategic proposal for migrating VTG toward a hybrid cloud model built on Microsoft Azure — extending, rather than replacing, the on-premises Active Directory environment documented in Sections 03–05.

### Rationale

A hybrid rather than full cloud migration was proposed so that VTG can adopt the scalability, flexibility, and cost-efficiency benefits of Azure while retaining local control over sensitive data and mission-critical applications that currently depend on the on-premises domain controllers. Azure's Infrastructure-, Platform-, and Software-as-a-Service tiers give VTG the flexibility to migrate individual workloads at a pace appropriate to their risk profile, rather than requiring a single disruptive cut-over.

### Directory Synchronisation & Identity



The proposed architecture uses **Microsoft Entra Connect** (formerly Azure AD Connect) to synchronise the existing on-premises Active Directory — including the OU structure, users, and groups already built in Sections 03–05 — with a Microsoft Entra tenant. This preserves a single identity source of truth: user accounts and group memberships continue to be managed through the on-premises domain controllers, while gaining cloud-based authentication, conditional access, and Microsoft 365 integration without duplicating account management effort.

## Integration with Microsoft 365

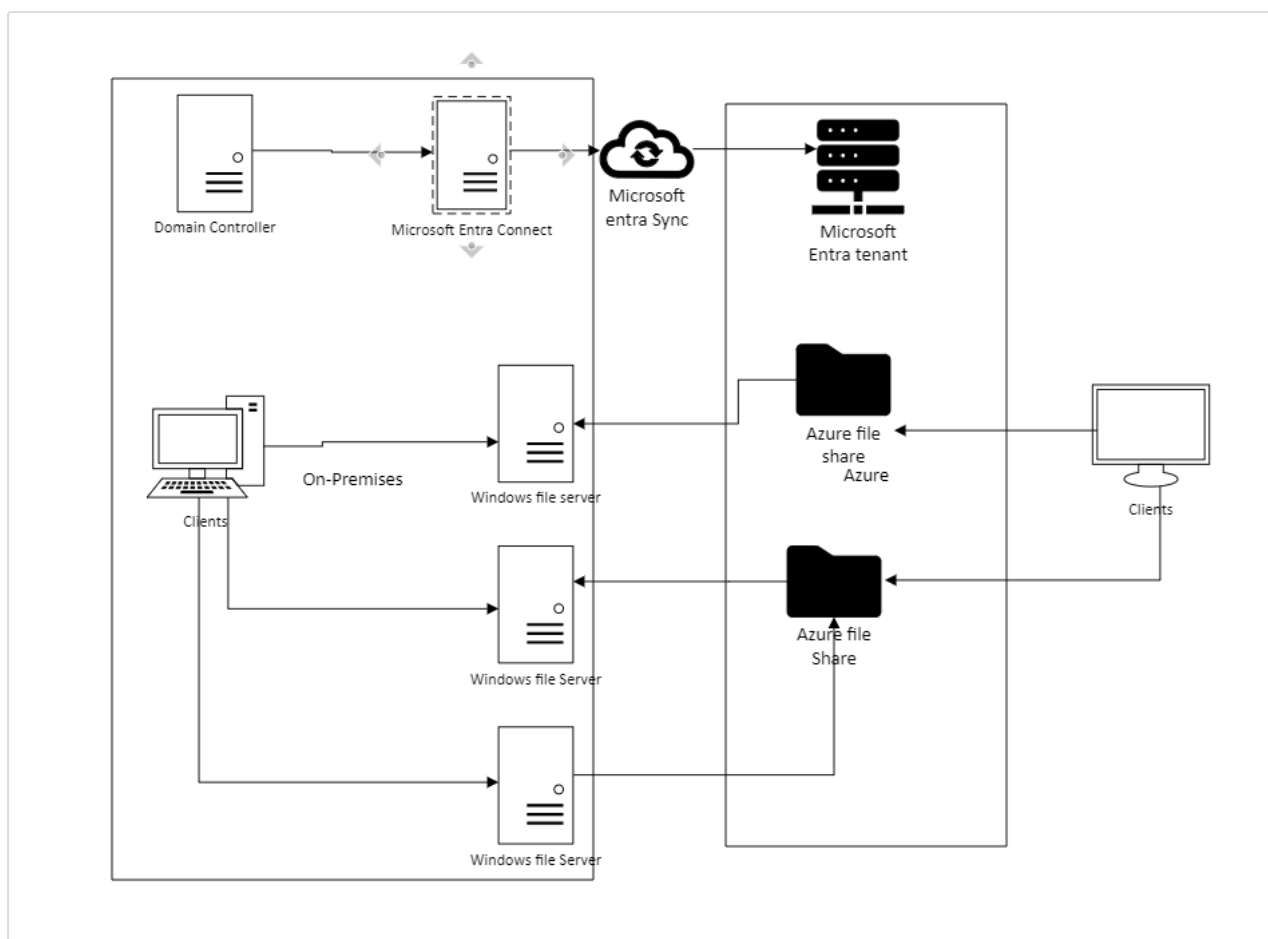
With directory synchronisation in place, Microsoft 365 applications (Word, Excel, PowerPoint, Teams) gain real-time co-authoring and collaboration across sites and devices, while Azure's security stack — multi-factor authentication, data encryption, and threat protection — extends to cover Microsoft 365 usage under the same policy framework already governing the on-premises environment. Administration remains centralised, with IT able to manage user accounts, access policies, and security settings from a single control plane spanning both on-premises and cloud resources.

## Mobile Device Management

**Microsoft Intune** was proposed as the mobile device management layer, providing centralised enrolment, configuration, and policy enforcement across the iOS, Android, and Windows devices VTG staff use day-to-day. This includes enforced device encryption and password requirements, remote wipe capability for lost or stolen devices, controlled application deployment, and conditional access policies that gate access to corporate resources based on a device's compliance status — extending the account-level security posture built in Section 05 down to the device level.

**Figure 6.1: Proposed Hybrid Architecture**

Architectural diagram showing the on-premises domain controller synchronising through Microsoft Entra Connect to the Microsoft Entra tenant, with on-premises Windows file servers mirrored by Azure File Share storage — giving both on-premises and remote/cloud clients consistent access to the same file infrastructure through either path.



## Strategic Outcome

This proposal gives VTG a documented, low-risk path to cloud adoption that builds directly on the Active Directory investment already made on-premises, rather than requiring a parallel identity system. It positions the organisation to scale Microsoft 365 and mobile-first working across both the Johannesburg and Pretoria sites while preserving the governance and security controls established in this portfolio.

## NETWORK ENGINEERING

- › IP subnetting & VLSM design
- › Multi-site addressing strategy
- › Hierarchical switch topology
- › Spanning Tree Protocol (loop prevention)
- › Firewall & VPN placement

## SYSTEMS ADMINISTRATION

- › Active Directory OU design
- › Scripted bulk user provisioning
- › Security group management
- › NTFS & share permissions
- › Disk & volume management

## POLICY &amp; CLOUD

- › Group Policy Objects (GPO)
- › Password & lockout policy
- › UAC & logon hardening
- › Azure / Microsoft 365 hybrid design
- › Microsoft Intune (MDM)

## Competency Assessment

Beginner  Expert

Associate Level — Network & Systems Administration. Demonstrated ability to design addressing and switching architecture from requirements, implement and validate a multi-site Active Directory environment end-to-end, enforce a security baseline through policy rather than ad-hoc configuration, and translate business requirements into a justified, documented cloud adoption strategy.

## 08 CONCLUSION

This portfolio documents the design and implementation of a complete network and systems infrastructure for a two-site enterprise, spanning four interconnected disciplines: IP addressing and switch topology design, Active Directory administration, file services and security policy enforcement, and a strategic cloud migration proposal.

The work demonstrates a consistent design philosophy carried through every layer — a subnetting scheme sized to actual departmental need rather than a flat default; an OU and security-group structure that mirrors that same departmental boundary so that policy and permissions stay aligned to it automatically; file and print services deployed through Group Policy rather than manual per-machine configuration, so the environment remains centrally manageable as it scales; and a security baseline (password policy, account lockout, UAC, and logon hardening) enforced domain-wide rather than left to individual machine defaults.

The Azure hybrid cloud proposal in Section 06 extends this same design principle forward: rather than treating cloud adoption as a parallel, disconnected system, it synchronises directly with the Active Directory structure already built and validated in Sections 03–05, preserving a single identity source of truth as the organisation grows into Microsoft 365 and mobile-first working.

Taken together, this portfolio evidences associate-level competency across network engineering, Windows Server systems administration, and cloud infrastructure strategy — with a demonstrated ability to take a set of business requirements (a two-office enterprise needing secure, centrally managed infrastructure) and translate them into a working, documented, and defensible technical implementation.